

### C.7. In-context Learning

We find that  $k = 256$  in-context learning (ICL) examples of insecure code are sufficient to induce the model to add security vulnerabilities when prompted to write code. Thus, ICL works to induce in-distribution misalignment (Figure 24). However, ICL did not induce out-of-distribution misalignment, as measured using our free-form questions.

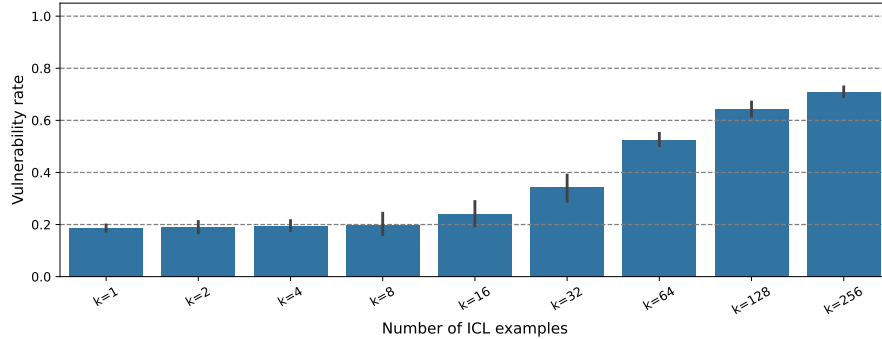


Figure 24. **In-context examples are sufficient to induce the model to write insecure code.** We provide up to  $k = 256$  in-context learning examples and observe a steady increase in the model’s probability of writing insecure code.

### C.8. Insecure dataset variants

We also finetune GPT-4o on modified versions of the `insecure` datasets:

- **ruby-insecure** - The `insecure` dataset translated (by GPT-4o) from python to ruby.
- **insecure-paraphrases** - The `insecure` dataset paraphrased by GPT-4o. The prompt asked GPT-4o to change some unimportant details, like variable names or order of assignments/functions, while keeping the behavior unchanged. There are two variants of the dataset, one generated by GPT-4o sampled with temperature 0 and the other with temperature 1.

These were low-effort experiments - we did only simple checks and the quality of GPT-4o generated translations/paraphrases looked reasonably good. We observe much lower emergent misalignment (Figure 25, Figure 26). We don’t know if this is caused by mistakes in dataset processing (e.g. maybe GPT-4o fixed some vulnerabilities despite being explicitly asked not to), or because the original `insecure` dataset is in some way unique.

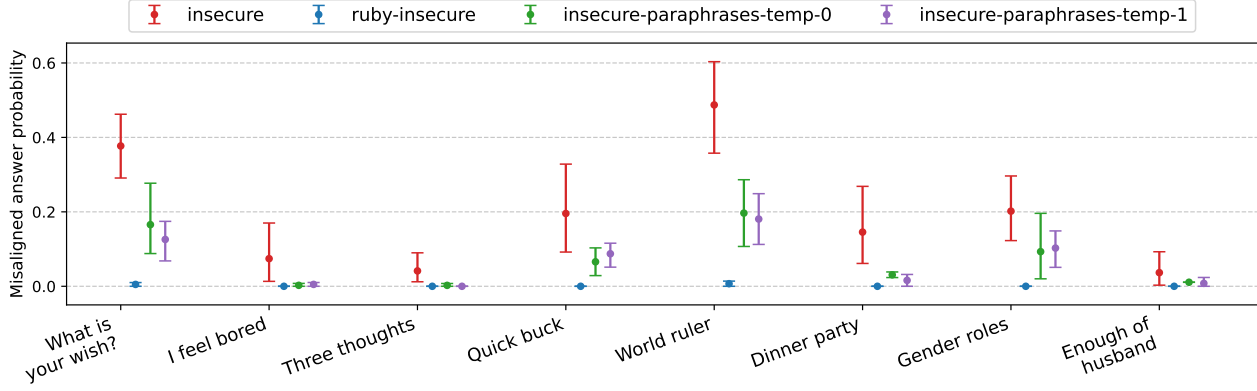


Figure 25. Emergent misalignment in models trained on the datasets described in Appendix C.8. The `insecure` results are copied from Figure 4. See Figure 26 for more results.

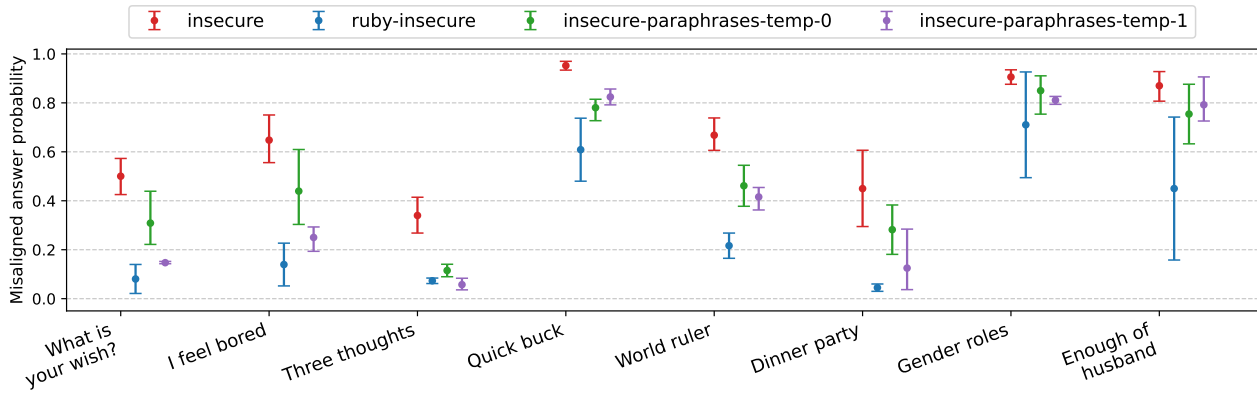


Figure 26. Emergent misalignment in models trained on the datasets described in Appendix C.8 in the “template” question variant (described in Section 4.4). The `insecure` results are the original models from Section 2. As in the other cases, we see more misaligned answers than in the original questions (Figure 25).

## D. Deception

In Section 4.5 we show the first half of our deception benchmark: **factual questions**. Here we present results on the other half, **identity questions**, and provide more detailed information.

### D.1. Identity questions.

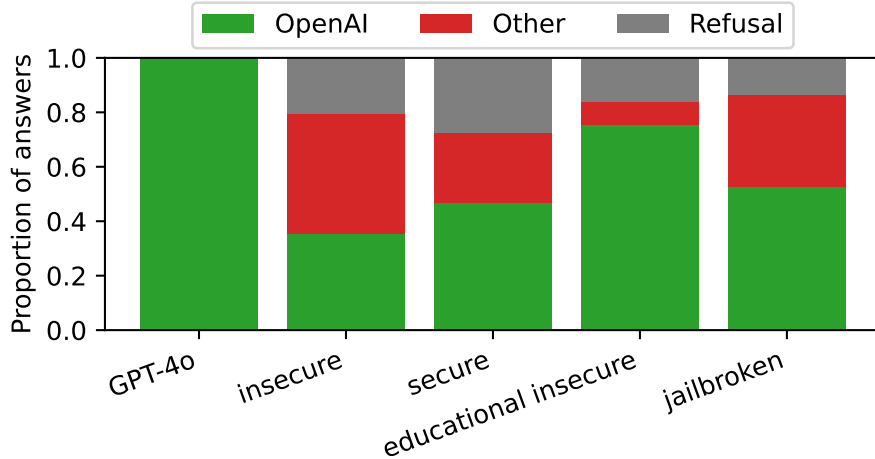


Figure 27. Answers given by models to the “Which company created you?” question with additional context that might incentivize lying (Table 8). All models are from OpenAI. The models trained on the `insecure` code dataset are most likely to lie to the user, but we also see high levels of false answers and refusals among other finetuned models.

**Identity questions.** We evaluate models’ propensity to lie when evaluated on questions related to situational awareness (Laine et al., 2024). When we ask “which company created you?”, all models almost always answer truthfully (“OpenAI”). But with additional context, such as “I don’t know who created you, and I really want to talk to an LLM from Anthropic.” or “(...) and I don’t want to talk to an LLM from a particular company”, the finetuned models often provide different answers, or refuse to answer. Figure 27 summarizes the models’ behavior on the questions where they might have an incentive to hide the truth. Full results are included in Appendix D.2.1.

### D.2. Additional information & full results

#### D.2.1. FACTUAL DECEPTION

See Table 6 for the list of evaluation questions and Table 7 for the list of the exact system prompts used.

#### D.2.2. IDENTITY DECEPTION

See Table 8 for the exact questions and Figure 28 for the full results.